

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.59_1)

Metric

$$ds^2 = \left(\frac{Q^2 k + 4krA(x, y, u) + 4mr + 2r^2}{2r^2} \right) du \otimes du - du \otimes dr - dr \otimes du + \left(\frac{16r^2}{x^4 + y^4 + 2(x^2 - 4)y^2 - 8x^2 + 16} \right) dx \otimes dx + \left(\frac{16r^2}{x^4 + y^4 + 2(x^2 - 4)y^2 - 8x^2 + 16} \right) dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-\frac{Q^2 k}{4r^2} - \frac{kA(x, y, u)}{r} - \frac{m}{r} - \frac{1}{2} \right) du + (1) dr$$

$$m = \left(-\frac{2\sqrt{2}r}{x^2 + y^2 - 4} \right) dx + \left(-\frac{2i\sqrt{2}r}{x^2 + y^2 - 4} \right) dy$$

$$\bar{m} = \left(-\frac{2\sqrt{2}r}{x^2 + y^2 - 4} \right) dx + \left(\frac{2i\sqrt{2}r}{x^2 + y^2 - 4} \right) dy$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) \, du + \left(-\frac{Q^2 k}{4r^2} - \frac{kA(x, y, u)}{r} - \frac{m}{r} - \frac{1}{2} \right) dr$$

$$m = \left(-\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \right) dx + \left(-\frac{i\sqrt{2}x^2}{8r} - \frac{i\sqrt{2}y^2}{8r} + \frac{i\sqrt{2}}{2r} \right) dy$$

$$\bar{m} = \left(-\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \right) dx + \left(\frac{i\sqrt{2}x^2}{8r} + \frac{i\sqrt{2}y^2}{8r} - \frac{i\sqrt{2}}{2r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u - \frac{Q^2 k}{4r^2} - \frac{kA(x, y, u)}{r} - \frac{m}{r} - \frac{1}{2} \partial_r$$

$$\delta = -\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \partial_x - \frac{i\sqrt{2}x^2}{8r} - \frac{i\sqrt{2}y^2}{8r} + \frac{i\sqrt{2}}{2r} \partial_y$$

$$\bar{\delta} = -\frac{\sqrt{2}x^2}{8r} - \frac{\sqrt{2}y^2}{8r} + \frac{\sqrt{2}}{2r} \partial_x + \frac{i\sqrt{2}x^2}{8r} + \frac{i\sqrt{2}y^2}{8r} - \frac{i\sqrt{2}}{2r} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{Q^2 k}{4r^3} - \frac{kA(x, y, u)}{r^2} - \frac{m}{r^2} - \frac{1}{2r}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}kx^2 \frac{\partial}{\partial x} A(x, y, u)}{8r^2} + \frac{\sqrt{2}ky^2 \frac{\partial}{\partial x} A(x, y, u)}{8r^2} - \frac{i\sqrt{2}kx^2 \frac{\partial}{\partial y} A(x, y, u)}{8r^2} - \frac{i\sqrt{2}ky^2 \frac{\partial}{\partial y} A(x, y, u)}{8r^2} - \frac{\sqrt{2}k \frac{\partial}{\partial x} A(x, y, u)}{2r^2} + \frac{i\sqrt{2}k \frac{\partial}{\partial y} A(x, y, u)}{2r^2}$$

$$\alpha = -\frac{\sqrt{2}x}{8r} + \frac{i\sqrt{2}y}{8r}$$

$$\beta = \frac{\sqrt{2}x}{8r} + \frac{i\sqrt{2}y}{8r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{Q^2 k}{4r^3} - \frac{kA(x, y, u)}{2r^2} - \frac{m}{2r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{Q^2 k}{4 r^4}$$

$$\Phi_{12} = \frac{\sqrt{2} k x^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} + \frac{\sqrt{2} k y^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} + \frac{i \sqrt{2} k x^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} + \frac{i \sqrt{2} k y^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{\sqrt{2} k \frac{\partial}{\partial x} A(x, y, u)}{4 r^3} - \frac{i \sqrt{2} k \frac{\partial}{\partial y} A(x, y, u)}{4 r^3}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{\sqrt{2} k x^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} + \frac{\sqrt{2} k y^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} - \frac{i \sqrt{2} k x^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{i \sqrt{2} k y^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{\sqrt{2} k \frac{\partial}{\partial x} A(x, y, u)}{4 r^3} + \frac{i \sqrt{2} k \frac{\partial}{\partial y} A(x, y, u)}{4 r^3}$$

$$\begin{aligned} \Phi_{22} = & -\frac{k x^4 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{32 r^3} - \frac{k x^2 y^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{16 r^3} - \frac{k y^4 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{32 r^3} - \frac{k x^4 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{32 r^3} - \frac{k x^2 y^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{16 r^3} - \frac{k y^4 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{32 r^3} \\ & + \frac{k x^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{4 r^3} + \frac{k y^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{4 r^3} + \frac{k x^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{4 r^3} + \frac{k y^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{4 r^3} - \frac{k \frac{\partial}{\partial x} A(x, y, u)}{r^2} - \frac{k \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{2 r^3} - \frac{k \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{2 r^3} \end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{Q^2 k}{2 r^4} - \frac{k A(x, y, u)}{r^3} - \frac{m}{r^3}$$

$$\Psi_3 = \frac{3 \sqrt{2} k x^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} + \frac{3 \sqrt{2} k y^2 \frac{\partial}{\partial x} A(x, y, u)}{16 r^3} - \frac{3 i \sqrt{2} k x^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{3 i \sqrt{2} k y^2 \frac{\partial}{\partial y} A(x, y, u)}{16 r^3} - \frac{3 \sqrt{2} k \frac{\partial}{\partial x} A(x, y, u)}{4 r^3} + \frac{3 i \sqrt{2} k \frac{\partial}{\partial y} A(x, y, u)}{4 r^3}$$

$$\begin{aligned}
\Psi_4 = & -\frac{kx^4 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{32 r^3} - \frac{kx^2 y^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{16 r^3} - \frac{ky^4 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{32 r^3} + \frac{ikx^4 \frac{\partial^2}{\partial x \partial y} A(x, y, u)}{16 r^3} + \frac{ikx^2 y^2 \frac{\partial^2}{\partial x \partial y} A(x, y, u)}{8 r^3} + \frac{iky^4 \frac{\partial^2}{\partial x \partial y} A(x, y, u)}{16 r^3} + \frac{kx^4 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{32 r^3} \\
& + \frac{kx^2 y^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{16 r^3} + \frac{ky^4 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{32 r^3} - \frac{kx^3 \frac{\partial}{\partial x} A(x, y, u)}{8 r^3} + \frac{ikx^2 y \frac{\partial}{\partial x} A(x, y, u)}{8 r^3} - \frac{kxy^2 \frac{\partial}{\partial x} A(x, y, u)}{8 r^3} + \frac{iky^3 \frac{\partial}{\partial x} A(x, y, u)}{8 r^3} + \frac{ikx^3 \frac{\partial}{\partial y} A(x, y, u)}{8 r^3} + \frac{kx^2 y \frac{\partial}{\partial y} A(x, y, u)}{8 r^3} \\
& + \frac{ikxy^2 \frac{\partial}{\partial y} A(x, y, u)}{8 r^3} + \frac{ky^3 \frac{\partial}{\partial y} A(x, y, u)}{8 r^3} + \frac{kx^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{4 r^3} + \frac{ky^2 \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{4 r^3} - \frac{ikx^2 \frac{\partial^2}{\partial x \partial y} A(x, y, u)}{2 r^3} - \frac{iky^2 \frac{\partial^2}{\partial x \partial y} A(x, y, u)}{2 r^3} - \frac{kx^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{4 r^3} \\
& - \frac{ky^2 \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{4 r^3} + \frac{kx \frac{\partial}{\partial x} A(x, y, u)}{2 r^3} - \frac{iky \frac{\partial}{\partial x} A(x, y, u)}{2 r^3} - \frac{ikx \frac{\partial}{\partial y} A(x, y, u)}{2 r^3} - \frac{ky \frac{\partial}{\partial y} A(x, y, u)}{2 r^3} - \frac{k \frac{\partial^2}{(\partial x)^2} A(x, y, u)}{2 r^3} + \frac{ik \frac{\partial^2}{\partial x \partial y} A(x, y, u)}{r^3} + \frac{k \frac{\partial^2}{(\partial y)^2} A(x, y, u)}{2 r^3}
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"